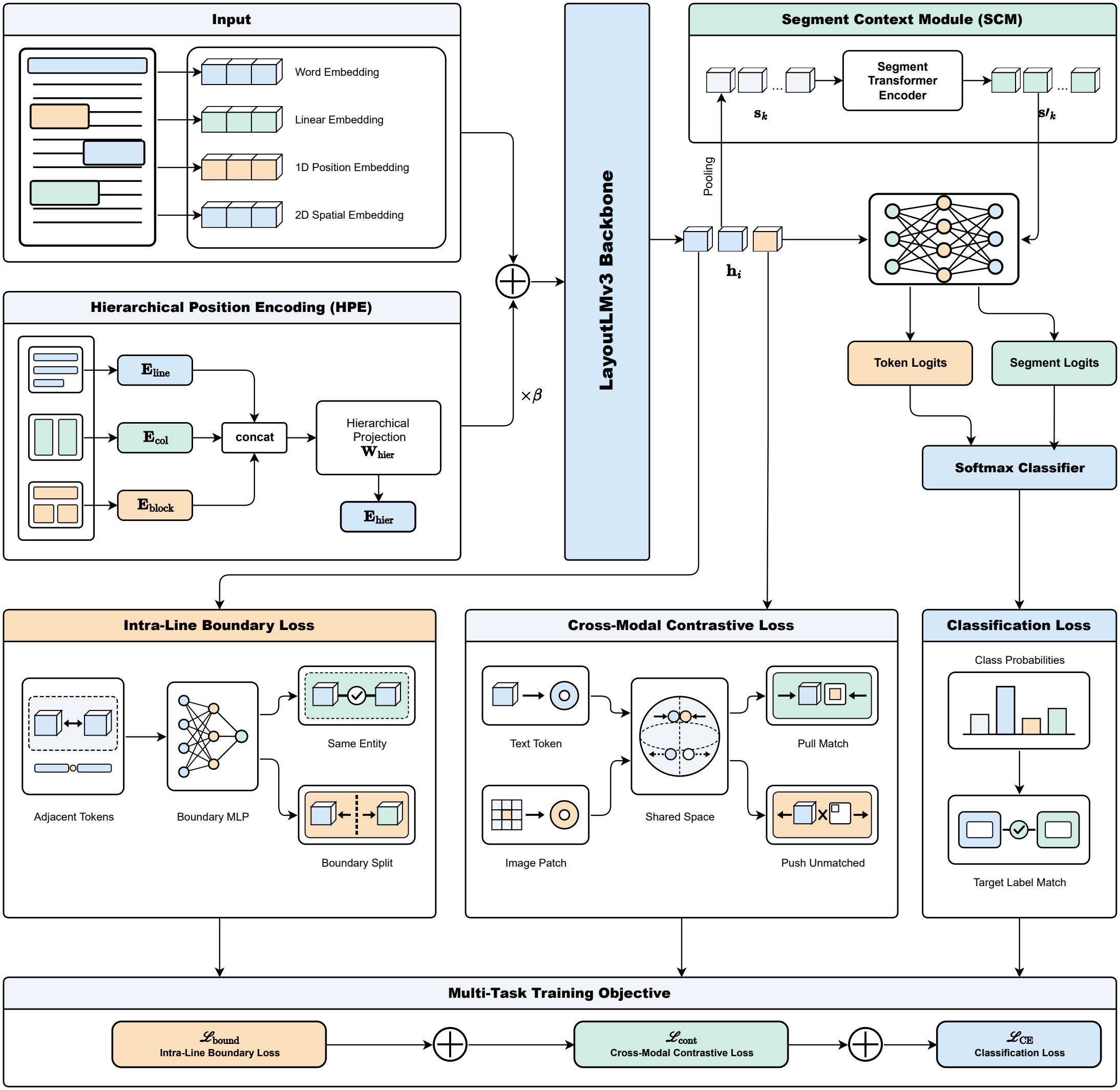

Input
Word Embedding
Linear Embedding
1D Position Embedding
2D Spatial Embedding
Hierarchical Position Encoding (HPE)
$\mathbf{E}_{\text{line}}$
$\mathbf{E}_{\text{col}}$
$\mathbf{E}_{\text{block}}$
concat
Hierarchical Projection $\mathbf{W}_{\text{hier}}$
$\mathbf{E}_{\text{hier}}$
$\times\beta$
LayoutLMv3 Backbone
Segment Context Module (SCM)
$\mathbf{s}_k$
Segment Transformer Encoder
$\mathbf{s}'_k$
Pooling
$\mathbf{h}_i$
Token Logits
Segment Logits
Softmax Classifier
Intra-Line Boundary Loss
Adjacent Tokens
Boundary MLP
Same Entity
Boundary Split
Cross-Modal Contrastive Loss
Text Token
Image Patch
Shared Space
Pull Match
Push Unmatched
Classification Loss
Class Probabilities
Target Label Match
Multi-Task Training Objective
$\mathscr{L}_{\text{bound}}$ Intra-Line Boundary Loss
$\mathscr{L}_{\text{cont}}$ Cross-Modal Contrastive Loss
$\mathscr{L}_{\text{CE}}$ Classification Loss